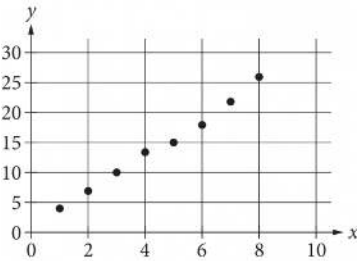


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Medium

Question



Which of the following could be the equation for a line of best fit for the data shown in the scatterplot above?

Answer

- A. $y = 3x + 0.8$
- B. $y = 0.8x + 3$
- C. $y = -0.8x + 3$
- D. $y = -3x + 0.8$

Correct Answer: A

Rationale

Choice A is correct. The data show a strong linear relationship between x and y. The line of best fit for a set of data is a linear equation that minimizes the distances from the data points to the line. An equation for the line of best fit can be written in slope-intercept form, $y = mx + b$, where m is the slope of the graph of the line and b is the y-coordinate of the y-intercept of the graph. Since, for the data shown, the y-values increase as the x-values increase, the slope of a line of best fit must be positive. The data shown lie almost in a line, so the slope can be roughly estimated using the formula for slope, $m = \frac{y_2 - y_1}{x_2 - x_1}$. The leftmost and rightmost data points have coordinates of about (1,4) and (8,26), so the slope is approximately $\frac{26 - 4}{8 - 1} = \frac{22}{7}$, which is a little greater than 3. Extension of the line to the left would intersect the y-axis at about (0,1). Only choice A represents a line with a slope close to 3 and a y-intercept close to (0,1).

Choice B is incorrect and may result from switching the slope and y-intercept. The line with a y-intercept of (0,3) and a slope of 0.8 is farther from the data points than the line with a slope of 3 and a y-intercept of (0,0.8). Choices C and D are incorrect. They represent lines with negative slopes, not positive slopes.